

KMJ DYNAMICS

AUTO-CART

The Autonomous Beach Cart — Complete Build Guide. The real working manual, expensive lessons included



● RC build — build only the green

● Autonomy — the full follow-me stack

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Free lifetime updates — this document improves as the cart does.

This is the free preview — every  block is real, tested content: the exact settings, part numbers, wiring rules, and the failures that produced them. Get the Complete Build Package at kmjdynamics.com

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- 1. Frame & chassis  [RC BUILD]
- 2. Steering  [RC BUILD]
- 3. Drivetrain (motors, chain, wheels)  [RC BUILD]
- 4. Power system & distribution  [RC BUILD — Buck #2 is  autonomy-only]
- 5. Compute (Raspberry Pi 5)  [AUTONOMY ONLY — the RC build has no Pi]
- 6. Motor controller (VESC Duet XS 60V dual VESC — VESC Labs)  [RC BUILD]
- 7. RC control  [RC BUILD — see the RC-only wiring box below]
- 8. Drive & steering software (drive.py / steering.py)  [AUTONOMY ONLY]
- 9. Sensors & telemetry  [AUTONOMY ONLY]
- 10. Displays (camera + 5" screen)  [AUTONOMY ONLY]
- 11. UWB localization — WORKING (4 anchors + tag, [4A] fix @3.3Hz)  [AUTONOMY ONLY]
- 12. Follow-me autonomy (modules/follow_me.py) — WORKING, road-tested  [AUTONOMY ONLY]
- 13. Safety systems (the stack, in order of what fires)  +  [VESC-side rows apply to RC too]
- 14. Networking, web panel & remote access  [AUTONOMY ONLY]
- 15. Software, running & autostart  [AUTONOMY ONLY]
- 16. 3D-printed parts (generators in 3d_models/, PETG unless noted)  +  [wheel hubs + battery box = RC too]
- 17. Key pin & config reference (full map: PINOUT.md)
- 18. Open items as of 8/3

Why this guide keeps getting better parts

I built this in layers on purpose. First a "dumb wagon" — a frame that rolls, steers, and drives by remote — and I made sure that worked before any self-driving stuff went on. Test each layer by itself before stacking the next one on top. That way, when something acts up, you know which layer to blame. I know the autonomous parts are the fun parts. But every hour I spent making a layer solid saved me five hours of head-scratching later. Trust me on this one.

Build the mechanical cart first (frame, steering, drivetrain), then power, then the computer and motor controller, then RC control, then sensors and displays, then the autonomy hardware (UWB) and the follow-me software, and finally the safety systems that wrap all of it.

0. Overview & build order

IN PLAIN ENGLISH

I built this in layers on purpose. First a "dumb wagon" — a frame that rolls, steers, and drives by remote — and I made sure that worked before any self-driving stuff went on. Test each layer by itself before stacking the next one on top. That way, when something acts up, you know which layer to blame. I know the autonomous parts are the fun parts. But every hour I spent making a layer solid saved me five hours of head-scratching later. Trust me on this one.

Build the mechanical cart first (frame, steering, drivetrain), then power, then the computer and motor controller, then RC control, then sensors and displays, then the autonomy hardware (UWB) and the follow-me software, and finally the safety systems that wrap all of it.

Cost and scope, up front: budget **\$3,500–4,100 in parts for the RC build**, or **\$4,300–5,100 for the full autonomous build** — the  autonomy stack is roughly \$800–1,000 on top of the  one. Where you land inside those ranges depends mostly on what your shop already stocks; STOCK_LIST.md breaks it down and names the two lines you can't shop your way out of. On top of the parts you need a 3D printer that can run carbon-fiber nylon and the ability to weld aluminum — if you have neither, that's another \$600–1,500 before you cut anything. **The RC build is the cheaper and simpler path** — build only the  chapters (§1–4, §6–7, plus the VESC-side rows of §13) and skip the entire lidar/UWB/sensor/compute stack; that's not how this cart was built, but those chapters stand alone, and the RC-only wiring box in §7 shows the simpler no-Pi hookup. **Two areas deliberately need your own ingenuity:** fitting the electronics into a weatherproof enclosure, and mounting everything to the frame to survive sand, salt, and vibration. The layout shown here is ONE working answer, not the only one — your parts and cable runs will differ.

/ **COLOR KEY (added 2026-08-09, launch prep):** every chapter here — and every section and line of STOCK_LIST.md — is tagged for which build actually needs it, because most builders only want the RC cart and should not overbuild for autonomy they aren't adding:

-  = **RC BUILD**. Required for the remote-control-only cart (the autonomous build uses all of this too — it's a superset). **Building RC? Buy and build ONLY the green items, then stop.**
-  = **AUTONOMY ONLY**. Only for the follow-me build. RC builders skip these chapters and parts entirely — this is the whole Pi/lidar/UWB/sensor stack, roughly \$800–1,000 of parts.
-  +  = **mixed** — mostly one, with the exceptions flagged inside the chapter.

(In the sellable PDF/HTML edition,  content renders with a green highlight and  with a purple one; in this Markdown master the dots carry the color.)

Since r3, STOCK_LIST.md additionally tags a handful of shop-consumable rows "**(OPTIONAL — shop stock)**" — wire, connector kits, heat shrink and the like that most shops already have. Use what you have; buy only what's missing. They stay  — the tier is about whether to BUY, not whether the build needs the function.

- 1.  Frame / chassis — 1.5" square aluminum, TIG welded, removable 1-1/4" side rails
 - 1.  Steering — wagon steering bracket + tie rods + 24V servo
 - 1.  Drivetrain — 2 BLDC motors, 5:1 chain, 16.5" wheels (BOND THE TIRES FIRST)
 - 1.  Power system — 24V 50Ah LiFePO4 (BMS-direct gauge), fuse, TVS, 5V buck, star ground (Buck #2 for the UWB bus = )
 - 1.  Compute — Raspberry Pi 5 + Argon NEO 5 + OS + cooling (RC build: no Pi at all)
 - 1.  Motor controller — VESC Duet XS 60V dual VESC (VESC Labs), configured over VESC Tool + CAN
 - 1.  RC control — DumboRC X4/X6F radio (RC-only builds wire it straight to the VESC/servo — see §7)
 - 1.  Drive & steering software — hardware PWM, ramps, auto park-brake, deadman
 - 1.  Sensors & telemetry — 6x TFmini ring, RPLIDAR S3, BMS-direct battery gauge over CAN/BLE
 - 1.  Displays — camera to the phone, radar on the 5" screen
 - 1.  UWB localization — 4 corner anchors + wearable tag (WORKING, 3.3Hz 4-anchor fix)
 - 1.  Follow-me autonomy — 4-corner solver + bearing PD steering (road-tested)
 - 1.  +  Safety systems — VESC-side failsafes are  for every build; the sensor gate stack is 
 - 1.  Networking, web panel & remote access
 - 1.  Software, autostart & backup
 - 1.  +  3D-printed parts — wheel hubs + battery box  ; sensor mounts, screen case, tag case 
 - 1. Pin & config reference (autonomous cart)
 - 1. Open items as of 7/11
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1. Frame & chassis ● [RC BUILD]

IN PLAIN ENGLISH

The frame is a flat rectangle of hollow square aluminum tube — picture a bed frame — 32 inches wide and 4 feet long, with extra tubes across it wherever something heavy sits (motors, axle, battery). I went with aluminum because it's light and won't rust at the beach, and welded because bolted joints shake loose in sand and vibration. Everything else in this guide bolts to this rectangle, so take your time getting it *square and flat*. A crooked frame means crooked wheels and a chain that keeps jumping off.

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№	Тема	Цели	Задачи	Результаты	Содержание
1	Введение	Понимать значение математики в жизни.	Знать основные понятия математики.	Уметь применять математические знания на практике.	Математика – это наука о числах, фигурах, измерениях. Математика – это язык, на котором говорят ученые. Математика – это искусство рассуждения. Математика – это способ мышления. Математика – это способ познания мира. Математика – это способ общения. Математика – это способ жизни.
2	Числа	Понимать значение чисел.	Знать основные свойства чисел.	Уметь применять знания о числах на практике.	Числа – это способ обозначения количества. Числа – это способ измерения. Числа – это способ сравнения. Числа – это способ упорядочивания. Числа – это способ классификации. Числа – это способ коммуникации. Числа – это способ мышления. Числа – это способ познания мира. Числа – это способ общения. Числа – это способ жизни.
3	Арифметика	Понимать значение арифметики.	Знать основные свойства арифметики.	Уметь применять знания об арифметике на практике.	Арифметика – это наука о числах. Арифметика – это искусство счета. Арифметика – это способ измерения. Арифметика – это способ сравнения. Арифметика – это способ упорядочивания. Арифметика – это способ классификации. Арифметика – это способ коммуникации. Арифметика – это способ мышления. Арифметика – это способ познания мира. Арифметика – это способ общения. Арифметика – это способ жизни.
4	Алгебра	Понимать значение алгебры.	Знать основные свойства алгебры.	Уметь применять знания об алгебре на практике.	Алгебра – это наука о уравнениях. Алгебра – это искусство решения задач. Алгебра – это способ измерения. Алгебра – это способ сравнения. Алгебра – это способ упорядочивания. Алгебра – это способ классификации. Алгебра – это способ коммуникации. Алгебра – это способ мышления. Алгебра – это способ познания мира. Алгебра – это способ общения. Алгебра – это способ жизни.
5	Геометрия	Понимать значение геометрии.	Знать основные свойства геометрии.	Уметь применять знания о геометрии на практике.	Геометрия – это наука о фигурах. Геометрия – это искусство измерения. Геометрия – это способ сравнения. Геометрия – это способ упорядочивания. Геометрия – это способ классификации. Геометрия – это способ коммуникации. Геометрия – это способ мышления. Геометрия – это способ познания мира. Геометрия – это способ общения. Геометрия – это способ жизни.
6	Тригонометрия	Понимать значение тригонометрии.	Знать основные свойства тригонометрии.	Уметь применять знания о тригонометрии на практике.	Тригонометрия – это наука о углах. Тригонометрия – это искусство измерения. Тригонометрия – это способ сравнения. Тригонометрия – это способ упорядочивания. Тригонометрия – это способ классификации. Тригонометрия – это способ коммуникации. Тригонометрия – это способ мышления. Тригонометрия – это способ познания мира. Тригонометрия – это способ общения. Тригонометрия – это способ жизни.
7	Аналитическая геометрия	Понимать значение аналитической геометрии.	Знать основные свойства аналитической геометрии.	Уметь применять знания об аналитической геометрии на практике.	Аналитическая геометрия – это наука о фигурах. Аналитическая геометрия – это искусство измерения. Аналитическая геометрия – это способ сравнения. Аналитическая геометрия – это способ упорядочивания. Аналитическая геометрия – это способ классификации. Аналитическая геометрия – это способ коммуникации. Аналитическая геометрия – это способ мышления. Аналитическая геометрия – это способ познания мира. Аналитическая геометрия – это способ общения. Аналитическая геометрия – это способ жизни.
8	Дифференциальное исчисление	Понимать значение дифференциального исчисления.	Знать основные свойства дифференциального исчисления.	Уметь применять знания о дифференциальном исчислении на практике.	Дифференциальное исчисление – это наука о функциях. Дифференциальное исчисление – это искусство измерения. Дифференциальное исчисление – это способ сравнения. Дифференциальное исчисление – это способ упорядочивания. Дифференциальное исчисление – это способ классификации. Дифференциальное исчисление – это способ коммуникации. Дифференциальное исчисление – это способ мышления. Дифференциальное исчисление – это способ познания мира. Дифференциальное исчисление – это способ общения. Дифференциальное исчисление – это способ жизни.
9	Интегральное исчисление	Понимать значение интегрального исчисления.	Знать основные свойства интегрального исчисления.	Уметь применять знания об интегральном исчислении на практике.	Интегральное исчисление – это наука о функциях. Интегральное исчисление – это искусство измерения. Интегральное исчисление – это способ сравнения. Интегральное исчисление – это способ упорядочивания. Интегральное исчисление – это способ классификации. Интегральное исчисление – это способ коммуникации. Интегральное исчисление – это способ мышления. Интегральное исчисление – это способ познания мира. Интегральное исчисление – это способ общения. Интегральное исчисление – это способ жизни.
10	Дополнительные задачи	Понимать значение дополнительных задач.	Знать основные свойства дополнительных задач.	Уметь применять знания о дополнительных задачах на практике.	Дополнительные задачи – это наука о функциях. Дополнительные задачи – это искусство измерения. Дополнительные задачи – это способ сравнения. Дополнительные задачи – это способ упорядочивания. Дополнительные задачи – это способ классификации. Дополнительные задачи – это способ коммуникации. Дополнительные задачи – это способ мышления. Дополнительные задачи – это способ познания мира. Дополнительные задачи – это способ общения. Дополнительные задачи – это способ жизни.

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5. [REDACTED], [REDACTED]. [REDACTED]
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7. [REDACTED] [REDACTED] [REDACTED] [REDACTED]. [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] (1/1" [REDACTED]
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[REDACTED] [REDACTED] ([REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] \$ [REDACTED], [REDACTED] [REDACTED]).

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IN PLAIN ENGLISH

[illegible]

1. $\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$. $\frac{1}{6} \cdot \frac{1}{4} = \frac{1}{24}$. $\frac{1}{24} \cdot \frac{1}{5} = \frac{1}{120}$. $\frac{1}{120} \cdot \frac{1}{6} = \frac{1}{720}$. $\frac{1}{720} \cdot \frac{1}{7} = \frac{1}{5040}$. $\frac{1}{5040} \cdot \frac{1}{8} = \frac{1}{40320}$. $\frac{1}{40320} \cdot \frac{1}{9} = \frac{1}{362880}$. $\frac{1}{362880} \cdot \frac{1}{10} = \frac{1}{3628800}$. $\frac{1}{3628800} \cdot \frac{1}{11} = \frac{1}{39916800}$. $\frac{1}{39916800} \cdot \frac{1}{12} = \frac{1}{479001600}$. $\frac{1}{479001600} \cdot \frac{1}{13} = \frac{1}{6227020800}$. $\frac{1}{6227020800} \cdot \frac{1}{14} = \frac{1}{87178291200}$. $\frac{1}{87178291200} \cdot \frac{1}{15} = \frac{1}{1307674368000}$. $\frac{1}{1307674368000} \cdot \frac{1}{16} = \frac{1}{20922790016000}$. $\frac{1}{20922790016000} \cdot \frac{1}{17} = \frac{1}{355687430272000}$. $\frac{1}{355687430272000} \cdot \frac{1}{18} = \frac{1}{6402373744992000}$. $\frac{1}{6402373744992000} \cdot \frac{1}{19} = \frac{1}{121645091154848000}$. $\frac{1}{121645091154848000} \cdot \frac{1}{20} = \frac{1}{2432901823096960000}$. $\frac{1}{2432901823096960000} \cdot \frac{1}{21} = \frac{1}{51090938285035136000}$. $\frac{1}{51090938285035136000} \cdot \frac{1}{22} = \frac{1}{1124000642280772992000}$. $\frac{1}{1124000642280772992000} \cdot \frac{1}{23} = \frac{1}{25852014772457778816000}$. $\frac{1}{25852014772457778816000} \cdot \frac{1}{24} = \frac{1}{620448354538986711680000}$. $\frac{1}{620448354538986711680000} \cdot \frac{1}{25} = \frac{1}{15511208863474667792000000}$. $\frac{1}{15511208863474667792000000} \cdot \frac{1}{26} = \frac{1}{403291430450341362592000000}$. $\frac{1}{403291430450341362592000000} \cdot \frac{1}{27} = \frac{1}{10888828622159216690976000000}$. $\frac{1}{10888828622159216690976000000} \cdot \frac{1}{28} = \frac{1}{304907201420458067347328000000}$. $\frac{1}{304907201420458067347328000000} \cdot \frac{1}{29} = \frac{1}{8842308841193284053072512000000}$. 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2. Steering 🟢 [RC BUILD]

IN PLAIN ENGLISH

The front wheels pivot on one swiveling beam, just like a toy wagon — actually, it *is* one. I salvaged the pivot assembly from a kid's wagon, and that solved the hardest mechanical problem for free. Instead of a kid pulling a handle, a strong servo motor swings the beam left and right through a "tie rod" (a threaded rod with a ball joint on each end, same idea as in your car). The Pi steers by sending the servo a timed electrical pulse; the servo moves to that angle and holds it. This part matters: the servo runs on 24 volts, and only its thin *signal* wire goes anywhere near the Pi.

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2019 年 12 月 31 日，公司应收账款账面余额为 1,000.00 万元，坏账准备为 100.00 万元，计提比例为 10.00%。2020 年 12 月 31 日，公司应收账款账面余额为 1,200.00 万元，坏账准备为 120.00 万元，计提比例为 10.00%。2021 年 12 月 31 日，公司应收账款账面余额为 1,500.00 万元，坏账准备为 150.00 万元，计提比例为 10.00%。2022 年 12 月 31 日，公司应收账款账面余额为 1,800.00 万元，坏账准备为 180.00 万元，计提比例为 10.00%。

(M): [REDACTED] [REDACTED]
[REDACTED] [REDACTED] [REDACTED] – [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED] [REDACTED].

[illegible]

1. 在 $\triangle ABC$ 中, $\angle C = 90^\circ$, $\angle A = 30^\circ$, $BC = 1$. 求 AB 和 AC 的长.

2. 在 $\triangle ABC$ 中, $\angle C = 90^\circ$, $\angle A = 45^\circ$, $AB = \sqrt{2}$. 求 AC 和 BC 的长.

3. 在 $\triangle ABC$ 中, $\angle C = 90^\circ$, $\angle A = 60^\circ$, $AB = 2$. 求 AC 和 BC 的长.

4. 在 $\triangle ABC$ 中, $\angle C = 90^\circ$, $\angle A = 30^\circ$, $AB = 2$. 求 AC 和 BC 的长.

5. 在 $\triangle ABC$ 中, $\angle C = 90^\circ$, $\angle A = 45^\circ$, $AB = \sqrt{2}$. 求 AC 和 BC 的长.

6. 在 $\triangle ABC$ 中, $\angle C = 90^\circ$, $\angle A = 60^\circ$, $AB = 2$. 求 AC 和 BC 的长.

7. 在 $\triangle ABC$ 中, $\angle C = 90^\circ$, $\angle A = 30^\circ$, $AB = 2$. 求 AC 和 BC 的长.

8. 在 $\triangle ABC$ 中, $\angle C = 90^\circ$, $\angle A = 45^\circ$, $AB = \sqrt{2}$. 求 AC 和 BC 的长.

9. 在 $\triangle ABC$ 中, $\angle C = 90^\circ$, $\angle A = 60^\circ$, $AB = 2$. 求 AC 和 BC 的长.

10. 在 $\triangle ABC$ 中, $\angle C = 90^\circ$, $\angle A = 30^\circ$, $AB = 2$. 求 AC 和 BC 的长.

1. 根据《中华人民共和国公司法》（以下简称《公司法》）第147条规定，股份有限公司的董事、监事、高级管理人员应当向公司申报其持有的本公司及与其有密切关联的其他公司的股份，不得转让其所持有的本公司股份。

2. 根据《上市公司收购管理办法》（以下简称《收购办法》）第14条规定，收购人通过协议收购方式取得一个上市公司已发行的有表决权股份达到5%时，应当在该事实发生之日起3日内，通知该上市公司，并公告该收购事实。

3. 根据《上市公司收购管理办法》第14条规定，收购人通过协议收购方式取得一个上市公司已发行的有表决权股份达到5%时，应当在该事实发生之日起3日内，通知该上市公司，并公告该收购事实。

4. 根据《上市公司收购管理办法》第14条规定，收购人通过协议收购方式取得一个上市公司已发行的有表决权股份达到5%时，应当在该事实发生之日起3日内，通知该上市公司，并公告该收购事实。

5. 根据《上市公司收购管理办法》第14条规定，收购人通过协议收购方式取得一个上市公司已发行的有表决权股份达到5%时，应当在该事实发生之日起3日内，通知该上市公司，并公告该收购事实。

6. 根据《上市公司收购管理办法》第14条规定，收购人通过协议收购方式取得一个上市公司已发行的有表决权股份达到5%时，应当在该事实发生之日起3日内，通知该上市公司，并公告该收购事实。

7. 根据《上市公司收购管理办法》第14条规定，收购人通过协议收购方式取得一个上市公司已发行的有表决权股份达到5%时，应当在该事实发生之日起3日内，通知该上市公司，并公告该收购事实。

8. 根据《上市公司收购管理办法》第14条规定，收购人通过协议收购方式取得一个上市公司已发行的有表决权股份达到5%时，应当在该事实发生之日起3日内，通知该上市公司，并公告该收购事实。

9. 根据《上市公司收购管理办法》第14条规定，收购人通过协议收购方式取得一个上市公司已发行的有表决权股份达到5%时，应当在该事实发生之日起3日内，通知该上市公司，并公告该收购事实。

10. 根据《上市公司收购管理办法》第14条规定，收购人通过协议收购方式取得一个上市公司已发行的有表决权股份达到5%时，应当在该事实发生之日起3日内，通知该上市公司，并公告该收购事实。

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4. Power system & distribution 🟢 [RC BUILD — Buck #2 is 🟡 autonomy-only]

IN PLAIN ENGLISH

Everything runs off one battery: a native 24-volt (25.6V nominal) 50Ah lithium iron phosphate pack. The big muscles (motors, steering servo) drink that "24V bus" straight; the delicate electronics get 5V from buck converters. Three things guard the bus: a **fuse** (melts and disconnects if a short tries to pull huge current — that's your fire protection), a **TVS diode** (a surge protector that clamps voltage spikes — I didn't know what a TVS diode was either, until a spike killed my first Raspberry Pi), and a **bulk capacitor** (a small energy sponge that smooths out jolts). And don't skip the "star ground" rule at the end: every ground wire meets at exactly one point. That keeps the motors' electrical noise out of the sensors' faint signals.

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1. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
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 $\frac{1}{8} \times \frac{1}{2} = \frac{1}{16}$
 $\frac{1}{16} \times \frac{1}{2} = \frac{1}{32}$
 $\frac{1}{32} \times \frac{1}{2} = \frac{1}{64}$
 $\frac{1}{64} \times \frac{1}{2} = \frac{1}{128}$
 $\frac{1}{128} \times \frac{1}{2} = \frac{1}{256}$
 $\frac{1}{256} \times \frac{1}{2} = \frac{1}{512}$
 $\frac{1}{512} \times \frac{1}{2} = \frac{1}{1024}$
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 $\frac{1}{4096} \times \frac{1}{2} = \frac{1}{8192}$
 $\frac{1}{8192} \times \frac{1}{2} = \frac{1}{16384}$
 $\frac{1}{16384} \times \frac{1}{2} = \frac{1}{32768}$
 $\frac{1}{32768} \times \frac{1}{2} = \frac{1}{65536}$
 $\frac{1}{65536} \times \frac{1}{2} = \frac{1}{131072}$
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5. Compute (Raspberry Pi 5) [AUTONOMY ONLY — the RC build has no Pi]

IN PLAIN ENGLISH

The Raspberry Pi is the cart's brain — an \$80 computer the size of a credit card, running Linux. Every sensor reading, steering decision, and safety check in this guide is a program running on it. Two things matter most right now: power it properly (clean 5V from Buck #1, never straight battery voltage) and keep it cool (a hot Pi slows itself down, and a slow brain makes a sluggish cart). The "hardware PWM" stuff below just means two of the Pi's pins can make perfectly steady timing pulses in dedicated silicon. Steering and throttle use those two pins because software-made pulses jitter — and jittery pulses made my motors twitch until I figured that out.

```

// Pin definitions for Raspberry Pi 5
// Motor pins
const int MOTOR_PIN_1 = 12; // PWM pin
const int MOTOR_PIN_2 = 13; // PWM pin
// Sensor pins
const int DISTANCE_PIN = 4; // Ultrasonic sensor
const int ANGLE_PIN = 5; // Gyro sensor
const int BATTERY_PIN = 6; // Battery voltage sensor

// Variables
float distance = 0;
float angle = 0;
float battery_voltage = 0;

// Functions
void setup() {
  pinMode(MOTOR_PIN_1, OUTPUT);
  pinMode(MOTOR_PIN_2, OUTPUT);
  pinMode(DISTANCE_PIN, INPUT);
  pinMode(ANGLE_PIN, INPUT);
  pinMode(BATTERY_PIN, INPUT);
}

void loop() {
  // Read sensor data
  distance = readDistance();
  angle = readAngle();
  battery_voltage = readBatteryVoltage();

  // Calculate steering and throttle
  float steering = calculateSteering(distance, angle);
  float throttle = calculateThrottle(battery_voltage);

  // Write motor pins
  digitalWrite(MOTOR_PIN_1, steering);
  digitalWrite(MOTOR_PIN_2, throttle);
}
```

1. `pinMode(MOTOR_PIN_1, OUTPUT);`
`pinMode(MOTOR_PIN_2, OUTPUT);`

2. `pinMode(DISTANCE_PIN, INPUT);`
`pinMode(ANGLE_PIN, INPUT);`
`pinMode(BATTERY_PIN, INPUT);`

3. `distance = readDistance();`
`angle = readAngle();`
`battery_voltage = readBatteryVoltage();`

4. `steering = calculateSteering(distance, angle);`
`throttle = calculateThrottle(battery_voltage);`

`digitalWrite(MOTOR_PIN_1, steering);`
`digitalWrite(MOTOR_PIN_2, throttle);`

IN PLAIN ENGLISH

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6b. Parking brake — OPTIONAL [RC BUILD]

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7. RC control 🟢 [RC BUILD — see the RC-only wiring box below]

IN PLAIN ENGLISH

This is a normal hobby-store radio set: a pistol-grip transmitter in your hand and a matchbox-sized receiver on the cart. The receiver puts out three "channels" — steering wheel, trigger, and a switch — each as a timed pulse the Pi reads on its pins. The cart is self-driving by default; flip the channel-3 switch and a human has the wheel instantly. One thing to catch: the receiver speaks in 5-volt pulses, and the Pi's pins can only take 3.3. Every signal goes through a small "level shifter" board that translates. Skip it and the Pi reads garbage — or worse, gets damaged. (Building the RC-only cart with no Pi? Then there's no shifter at all — see the green box below.)

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- $\mathbf{r}_i = \mathbf{r}_i^0 + \mathbf{v}_i^0 t + \frac{1}{2} \mathbf{a}_i^0 t^2$ (Eq. 1). $\mathbf{r}_i^0$ and $\mathbf{v}_i^0$ are the initial position and velocity of the particle i . $\mathbf{a}_i^0$ is the initial acceleration of the particle i .
- $\mathbf{F}_i = -\nabla V(\mathbf{r}_i)$ (Eq. 2): $\mathbf{F}_i$ is the force acting on the particle i , $V(\mathbf{r}_i)$ is the potential energy of the particle i . $\mathbf{F}_i$ is calculated using the force field parameters (FFPs) and the coordinates of the particles.
- $\mathbf{v}_i = \mathbf{v}_i^0 + \mathbf{a}_i^0 t$ (Eq. 3): $\mathbf{v}_i$ is the velocity of the particle i , $\mathbf{v}_i^0$ is the initial velocity of the particle i , $\mathbf{a}_i^0$ is the initial acceleration of the particle i . $\mathbf{v}_i$ is calculated using the force $\mathbf{F}_i$ and the mass m_i of the particle i .
- $\mathbf{r}_i = \mathbf{r}_i^0 + \mathbf{v}_i^0 t + \frac{1}{2} \mathbf{a}_i^0 t^2$ (Eq. 4): $\mathbf{r}_i$ is the position of the particle i , $\mathbf{r}_i^0$ is the initial position of the particle i , $\mathbf{v}_i^0$ is the initial velocity of the particle i , $\mathbf{a}_i^0$ is the initial acceleration of the particle i . $\mathbf{r}_i$ is calculated using the force $\mathbf{F}_i$ and the mass m_i of the particle i .

8. Drive & steering software (drive.py / steering.py) 🟡 [AUTONOMY ONLY]

- $\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{d}{dt} \left(\frac{1}{2} m \left(\frac{dx}{dt} \right)^2 \right) = m v \frac{dv}{dt} = m v a$, where a is the acceleration. This is the rate of change of kinetic energy.
- The work done by a force F over a distance s is $W = \int F ds$. If the force is constant, $W = F s$. If the force varies, the work is the area under the $F-s$ graph.

- $\frac{1}{\sqrt{1-\beta^2}} = \gamma$ (Lorentz factor): $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ (where $\beta = \frac{v}{c}$). $\Delta t = \gamma \Delta t_0$ (time dilation). $\Delta x = \gamma \Delta x_0$ (length contraction).
- $\frac{1}{\sqrt{1-\beta^2}} = \gamma$ (Lorentz factor): $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ (where $\beta = \frac{v}{c}$). $\Delta t = \gamma \Delta t_0$ (time dilation). $\Delta x = \gamma \Delta x_0$ (length contraction).
- $\frac{1}{\sqrt{1-\beta^2}} = \gamma$ (Lorentz factor): $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ (where $\beta = \frac{v}{c}$). $\Delta t = \gamma \Delta t_0$ (time dilation). $\Delta x = \gamma \Delta x_0$ (length contraction).
- $\frac{1}{\sqrt{1-\beta^2}} = \gamma$ (Lorentz factor): $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ (where $\beta = \frac{v}{c}$). $\Delta t = \gamma \Delta t_0$ (time dilation). $\Delta x = \gamma \Delta x_0$ (length contraction).
- $\frac{1}{\sqrt{1-\beta^2}} = \gamma$ (Lorentz factor): $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ (where $\beta = \frac{v}{c}$). $\Delta t = \gamma \Delta t_0$ (time dilation). $\Delta x = \gamma \Delta x_0$ (length contraction).
- $\frac{1}{\sqrt{1-\beta^2}} = \gamma$ (Lorentz factor): $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ (where $\beta = \frac{v}{c}$). $\Delta t = \gamma \Delta t_0$ (time dilation). $\Delta x = \gamma \Delta x_0$ (length contraction).
- $\frac{1}{\sqrt{1-\beta^2}} = \gamma$ (Lorentz factor): $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ (where $\beta = \frac{v}{c}$). $\Delta t = \gamma \Delta t_0$ (time dilation). $\Delta x = \gamma \Delta x_0$ (length contraction).
- $\frac{1}{\sqrt{1-\beta^2}} = \gamma$ (Lorentz factor): $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ (where $\beta = \frac{v}{c}$). $\Delta t = \gamma \Delta t_0$ (time dilation). $\Delta x = \gamma \Delta x_0$ (length contraction).

9. Sensors & telemetry [AUTONOMY ONLY]

IN PLAIN ENGLISH

Sensors are the cart's senses. This section wires up three of them: six little laser rangefinders that watch for obstacles, a battery gauge that works like a smart fuel gauge, and a spinning laser that maps everything around the cart. Each subsection's Overview box explains its own sensor.

IN PLAIN ENGLISH

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IN PLAIN ENGLISH

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10. Displays (camera + 5" screen) 🟡 [AUTONOMY ONLY]

IN PLAIN ENGLISH

The cart has two "faces": your phone gets the live camera feed plus every control and gauge through a web page (§14), and a small 5-inch screen on the cart shows a radar-style sweep of what the 360° lidar sees. The radar is drawn mirror-style on purpose — forward points down. During follow-me you're standing in *front* of the cart looking back at the screen, so this way left is left and right is right from where you stand.

$\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \mathbf{F} \cdot \mathbf{v}$; $\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \mathbf{F} \cdot \mathbf{v}$; $\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \mathbf{F} \cdot \mathbf{v}$; $\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \mathbf{F} \cdot \mathbf{v}$.

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11. UWB localization — WORKING (4 anchors + tag, [4A] fix @3.3Hz)

● [AUTONOMY ONLY]

IN PLAIN ENGLISH

This is how the cart knows where you are — the heart of follow-me. UWB (ultra-wideband) is basically indoor GPS: four radio "anchors" on the cart's corners each ping a small "tag" you carry and time how long the radio wave takes to come back. Radio travels about a foot per nanosecond, so the round-trip time *is* the distance. Four distances from four known corners pin down exactly where the tag is, same as GPS satellites pin down your phone. This section has the strictest rules in the whole guide, because these little radio modules are the most fragile thing on the cart. They run on exactly 3.3 volts and die instantly at 5. They'll even kill *themselves* if powered without two tiny "decoupling" capacitors steadying their power during a transmission burst. I didn't know any of that going in — about half the modules I ever bought died teaching me. Follow the rules below and yours won't.

$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = \frac{\partial L}{\partial x}$

where L is the Lagrangian function, defined as the difference between the kinetic energy (T) and the potential energy (V):

$$L = T - V$$

The kinetic energy T is given by:

$$T = \frac{1}{2} m \dot{x}^2$$

and the potential energy V is given by:

$$V = \frac{1}{2} k x^2$$

where m is the mass of the object, k is the spring constant, and x is the displacement from the equilibrium position.

$$\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(x) e^{-x^2} dx = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(x) e^{-x^2} dx$$

- [illegible]

$$\frac{1}{2} \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{1}{2}$$
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2. $\frac{1}{x^2} = x^{-2}$, $\frac{d}{dx} x^{-2} = -2x^{-3} = -\frac{2}{x^3}$. $\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$. $\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$.
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```

def follow_me():
    # Get the four anchor distances
    d1 = get_distance(ANCHOR1)
    d2 = get_distance(ANCHOR2)
    d3 = get_distance(ANCHOR3)
    d4 = get_distance(ANCHOR4)

    # Calculate the position (x, y) using trilateration
    (x, y) = trilaterate(d1, d2, d3, d4)

    # Calculate the distance to the robot
    d_robot = get_distance(ROBOT)

    # Calculate the angle to the robot
    angle = get_angle(ROBOT)

    # Calculate the steering angle
    steering_angle = angle - math.atan2(y - ROBOT_Y, x - ROBOT_X)

    # Calculate the speed
    speed = 1.0 / (1.0 + d_robot)

    # Drive the robot
    drive(steering_angle, speed)

```

12. Follow-me autonomy (modules/follow_me.py) — WORKING, road-tested 🟡 [AUTONOMY ONLY]

IN PLAIN ENGLISH

This is the program that does the actual following. Every third of a second it takes the four anchor distances, works out where you are (a bit of geometry called "trilateration" — finding the one point that fits all four measured distances), and turns that into two numbers: which way to steer (toward you) and how fast to go (based on how far away you are — it slows as it gets close and stops about 8 feet away, so it never crowds you). On top of that are the manners that make it trustworthy: it wakes up disarmed and won't move until you deliberately cycle the tag off and on, it refuses to drive at you if you're behind it, and it never, ever reverses on its own in sand — a stuck cart has to stay light enough to push out by hand.

```

def follow_me():
    # Get the four anchor distances
    d1 = get_distance(ANCHOR1)
    d2 = get_distance(ANCHOR2)
    d3 = get_distance(ANCHOR3)
    d4 = get_distance(ANCHOR4)

    # Calculate the position (x, y) using trilateration
    (x, y) = trilaterate(d1, d2, d3, d4)

    # Calculate the distance to the robot
    d_robot = get_distance(ROBOT)

    # Calculate the angle to the robot
    angle = get_angle(ROBOT)

    # Calculate the steering angle
    steering_angle = angle - math.atan2(y - ROBOT_Y, x - ROBOT_X)

    # Calculate the speed
    speed = 1.0 / (1.0 + d_robot)

    # Drive the robot
    drive(steering_angle, speed)

    # Disarm the robot
    disarm()

    # Wait for the tag to be cycled on and off
    while True:
        if get_tag_state() == TAG_ON:
            # Tag is on, start following
            follow_me()
        elif get_tag_state() == TAG_OFF:
            # Tag is off, stop following
            stop()

```

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IN PLAIN ENGLISH

Safety here is an onion, not one switch: a stack of separate layers — eleven at last count, each able to stop the cart on its own. They run from "sees a problem coming" (the proximity gate) down to "everything crashed" (a script that parks the wheels even if the whole program dies). The rule behind all of them is simple: any single failure should make the cart *stop*, never make it run away. The table reads top to bottom in the order things would fire. And check the gap list at the bottom — I keep an honest list of what's *not* protected yet, because pretending it's all covered is how you get surprised.

 	
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[illegible]

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15. Software, running & autostart 🟡 [AUTONOMY ONLY]

IN PLAIN ENGLISH

The cart's software starts itself: flip the power on, and about a minute later everything — drive, sensors, follow-me, the radar screen, the web panel — is running with no keyboard or monitor. That's done with "services" (Linux for "always run this program, restart it if it dies, run this cleanup if it crashes"). This section lists them all, plus how to run things by hand when you're tinkering, and the automatic backups that push every change to a private online copy twice an hour.

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[illegible]

17. Key pin & config reference (full map: PINOUT.md)

IN PLAIN ENGLISH

The one-page cheat sheet: which wire lands on which Pi pin, plus the key electrical numbers. When something's acting up or you're rebuilding, check this list first.

- [illegible]

18. Open items as of 8/3

IN PLAIN ENGLISH

The honest to-do list. Every real build has one — putting it out here keeps me from pretending the cart is more finished than it is.

- [illegible]

This is the free preview — every  block is real, tested content: the exact settings, part numbers, wiring rules, and the failures that produced them. Get the Complete Build Package at kmdynamics.com

kmjdynamics.com